Concavity, Curve Sketching, and Optimization

Section 1: Concavity & Points of Inflection

1 On which interval is the function $f(x) = x^3 - 6x^2 + 9x$ concave up?

a. $(2, \infty)$

b. $(-\infty, 2)$

c. $(1, 3)$

d. $(0, \infty)$

2 What is the point of inflection of the curve $f(x) = x^3 - 6x^2 + 9x$?

a. $(2, 2)$

b. $(2, 0)$

c. $(1, 4)$

d. $(3, 0)$

3 What is a necessary condition for $(c, f(c))$ to be a point of inflection?

a. $f''(c) = 0$ or $f''(c)$ undefined, AND f'' changes sign across c

b. $f'(c) = 0$

c. $f''(c) = 0$ only

d. $f(c) = 0$

Section 2: Second Derivative Test for Local Extrema

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4 If $f'(3) = 0$ and $f''(3) = -4$, what does the Second Derivative Test conclude about $x = 3$?

a. f has a local maximum at $x = 3$

b. f has a local minimum at $x = 3$

c. The test is inconclusive

d. $x = 3$ is an inflection point

5 When is the Second Derivative Test inconclusive at a critical point c?

a. When $f''(c) = 0$ or $f''(c)$ does not exist

b. When $f''(c) > 0$

c. When $f'(c) = 0$

d. When $f(c) = 0$

6 Apply the Second Derivative Test to $f(x) = x^4 - 4x^3$ at its critical point $x = 3$.

a. Local minimum because $f''(3) = 36 > 0$

b. Local maximum because $f''(3) < 0$

c. Inflection point

d. Inconclusive

Section 3: Applied Optimization: Geometric Problems

7 A farmer has 200 meters of fencing to enclose a rectangular field adjacent to a straight river (no fence needed along the river). What dimensions maximize the area?

a. Length 100 m, Width 50 m

b. Length 80 m, Width 60 m

c. Length 120 m, Width 40 m

d. Square of side 66.7 m

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8 An open-top rectangular box is formed by cutting equal squares of side x from the corners of a 12 by 12 cm sheet of cardboard. What value of x maximizes the box volume?

a. $x = 2$ cm

b. $x = 3$ cm

c. $x = 1.5$ cm

d. $x = 4$ cm

9 What is the maximum volume in the box problem above?

a. $128\ \text{cm}^3$

b. $144\ \text{cm}^3$

c. $108\ \text{cm}^3$

d. $96\ \text{cm}^3$

Section 4: Applied Optimization: Cost & Distance

10 Which point on the line $y = 2x + 1$ is closest to the origin $(0, 0)$?

a. $(-\frac{2}{5}, \frac{1}{5})$

b. $(-1, -1)$

c. $(0, 1)$

d. $(-\frac{1}{2}, 0)$

11 A company determines that total cost for producing x units is $C(x) = 0.01x^2 + 4x + 100$. What production level minimizes average cost $\bar{C}(x) = \frac{C(x)}{x}$?

a. $x = 100$

b. $x = 50$

c. $x = 200$

d. $x = 400$

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12 Two positive numbers have a product of 36. What is their minimum sum?

a. 12

b. 13

c. 15

d. 20

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